



UGen300 Series

Portable on-device AI for
classic AI and generative AI workloads.



UGen300
M.2 AI Module **2** GB

UGen300
M.2 AI Module **8** GB



UGen300
USB AI Accelerator **8** GB

RUN AI YOUR WAY

Local AI • Secure • Fast • Scalable

Clean • Premium • AI-Ready ⁸



UGen300 Platform — Hailo-10H

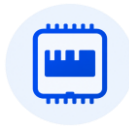
Up to 40 TOPS (INT4) at 2.5 W typical — supports both classic AI and generative AI.



Chipset - Hailo-10H + LPDDR4/4X

Delivers up to 40 TOPS (INT4) at 2.5W (typical), support both classic AI and generative AI.

Memory

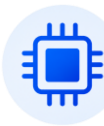


LPDDR4/4X — High-bandwidth memory for seamless AI.

Supported Operating System



Windows, Linux and Android supported.



Host Architecture

x86 and ARM host processors.



Supported Models

200+ pre-trained models via the online model zoo including **vision** network and **generative AI** models.

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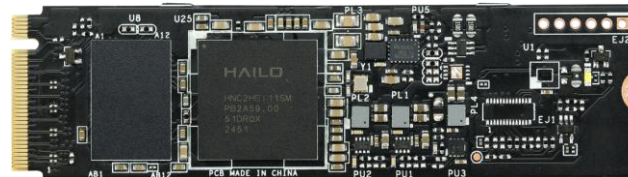
UGen300 Series Spec.



UGen300 USB AI Accelerator

	Memory	LPDDR4 8GB
	Interface	USB Type-C
	Dimension	105 x 50 x 18 mm
	Weight	< 150 g

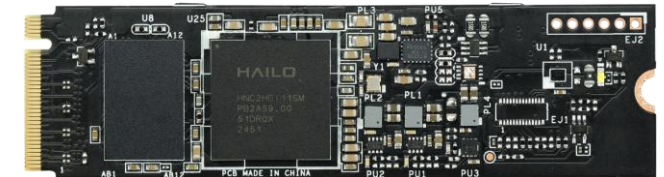
Compact USB-C accessory bringing [Classic AI](#) and [GenAI inference](#) to existing systems.



UGen300 M2 AI Module 2GB

	Memory	LPDDR4X 2GB
	Interface	PCIe M.2 Key-M
	Dimension	22 x 80 x 1 mm
	Weight	6 grams

For compact systems — optimized for [vision](#) models and small language models ([SLMs](#)).



UGen300 M2 AI Module 8GB

	Memory	LPDDR4 8GB
	Interface	PCIe M.2 Key-M
	Dimension	22 x 80 x 1 mm
	Weight	6 grams

For compact systems — optimized for [vision](#) and [GenAI](#) models (LLM, VLM).

UGen Series — Model Coverage & Use Cases

From pure vision to generative AI — 218 models, every SKU mapped to real, runnable demos.

218

models mapped

across 4 SKUs

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What UGen Series Can Do ...

GenAI • VLM • Vision AI • Edge Deployment



LLM

Explain edge AI in simple terms.

Edge AI runs AI models locally on devices, enabling real-time inference with lower latency and privacy.



VLM

Ask questions about any image.

AI answers from image content — fully on-device.



Detection

Detect and locate objects in real time.

Objects are detected live with labels and confidence.



Classification

Identify and classify objects accurately.

This image contains a Dog and a Cat.



Segmentation

Understand regions and boundaries in the scene.

The sofa, pillows, table, and plant are segmented.



Pose

Analyze posture and movement in real time.

Pose scoring with real-time coaching feedback.



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206 Vision Models Across 27 Tasks — All 4 SKUs

Every SKU (including UGen200) runs the full vision zoo. Numbers = models per task.

61	Object Detection	53	Image Classification	15	Text-Image Retrieval	13	Zero-shot Classification
16	Instance Segmentation	6	Semantic Segmentation	5	Face Detection	4	Oriented-Box (OBB)
4	Super-Resolution	3	Multi-person Pose	3	Single-person Pose	2	Face Recognition
2	Facial Landmarks	2	Person Re-ID	2	Depth Estimation	2	Zero-shot Depth
2	Denoising	2	Low-light Enhance	9	+ 9 more tasks ×1		

206 Vision + 12 GenAI = 218 total on Hailo-10H. All 206 vision models run on UGen200 too.

12 GenAI Models Mapped to Every SKU

LLM ×7 · VLM ×2 · Whisper ×3 — none run on UGen200 (Hailo-8L is vision-only).

Model	Type	Params	200	2G	8G	USB
Qwen3-1.7B-Instruct	LLM	1.7B	—	—	✓	✓
Qwen2.5-1.5B-Instruct	LLM	1.5B	—	△	✓	✓
Qwen2.5-Coder-1.5B-Instruct	LLM	1.5B	—	△	✓	✓
Qwen2-1.5B-Instruct	LLM	1.5B	—	△	✓	✓
Qwen2-1.5B-Instruct-Function-Calling-v1	LLM	1.5B	—	△	✓	✓
Llama3.2-1B-Instruct	LLM	1B	—	✓	✓	✓
DeepSeek-R1-Distill-Qwen-1.5B	LLM	1.5B	—	△	✓	✓
Qwen3-VL-2B-Instruct	VLM	2B	—	—	✓	✓
Qwen2-VL-2B-Instruct	VLM	2B	—	—	✓	✓
Whisper-Tiny	ASR	~39M	—	✓	✓	✓
Whisper-Base	ASR	74M	—	✓	✓	✓
Whisper-Small	ASR	~244M	—	✓	✓	✓

△ = 1.5B/1.7B need INT4 on 2G and are memory-tight; official 2G positioning is "LLM ~1B". Source: Hailo GenAI Model Zoo.

7 LLM

1B–1.7B, all on 8G/USB

2 VLM

2B — 8G/USB only

3 Whisper

runs from 2G up

GenAI per SKU

200: 0 2G: 4

8G / USB: 12

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UGen Series Model Coverage at a Glance

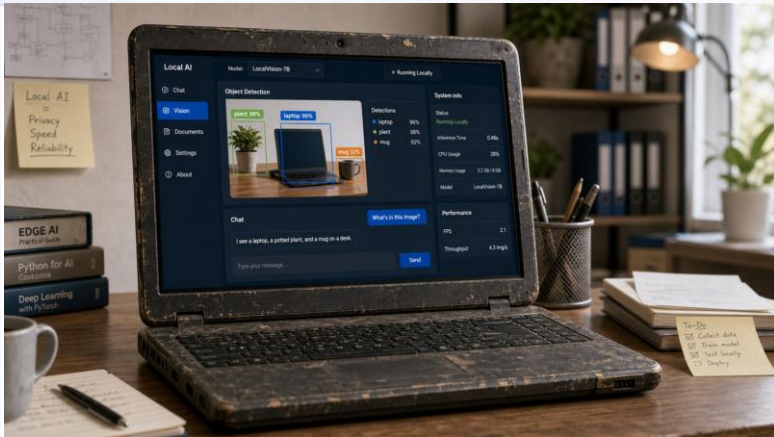
One product line — from 206 pure-vision models to the full 218 with generative AI.

Capability	UGen200	UGen300 M.2 2G	UGen300 M.2 8G	UGen300 USB 8G
Chipset	Hailo-8L	Hailo-10H	Hailo-10H	Hailo-10H
Vision models (206 / 27 tasks)	✓	✓	✓	✓
Whisper speech (3)	—	✓	✓	✓
LLM (7)	—	△	✓	✓
VLM (2)	—	—	✓	✓
Total runnable models	206	≈210	218	218

Takeaway: 8G / USB run all 218 models. UGen200 owns 206 pure-vision at the lowest cost. 2G sits in between (adds voice + light LLM).

Local AI Anywhere

Offline, private, low-cost - revive old hardware, teach coding, run off-grid.



Revive Old Devices

Give aging laptops & PCs modern local AI - no new machine.



Coding & Cram Schools

Hands-on AI/coding lessons - private, offline, low-cost.



Off-grid & Remote

Full AI where there is no network at all.

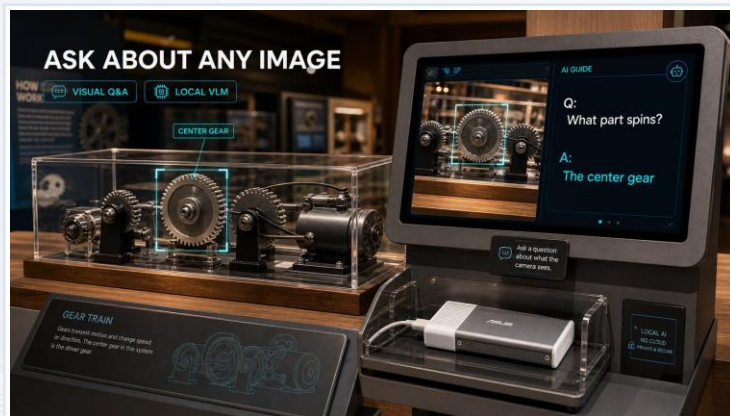
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UGen300 8G / USB See & Understand — VLM

2B vision-language models unlock look-and-answer. 8GB only.



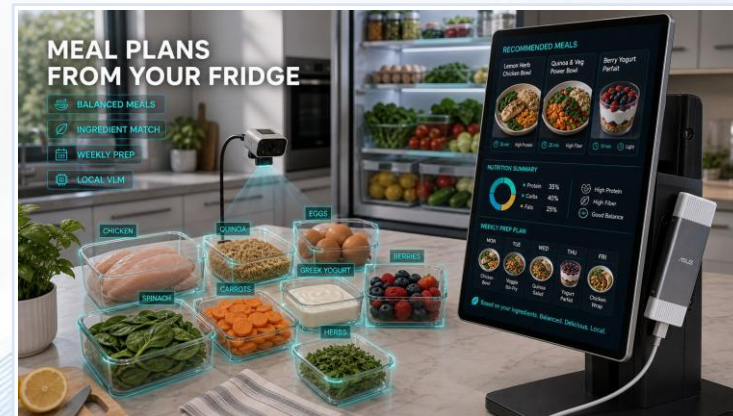
V1 · Ask About Image

github.com/erp0917-stack/ugen300-demos



V2 · Outfit Advisor

github.com/erp0917-stack/ugen300-demos



V3 · Fridge Chef

github.com/erp0917-stack/ugen300-demos



FACE1 · Face Report

github.com/erp0917-stack/ugen300-demos



OpenClaw Remote Monitor

github.com/Asus-UGen-Series/ugen300-openclaw-edge-remote-monitor

Why 8G?

Qwen2-VL-2B is a full 2B VLM — beyond the 2G “small-VLM” tier. This is your reason to move customers to 8G.

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Source Code → **GitHub**



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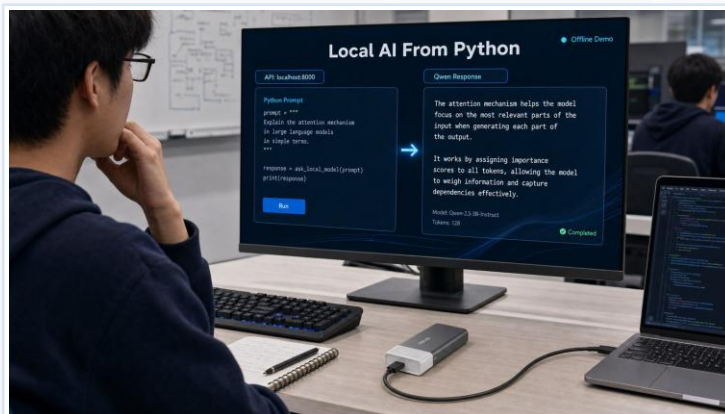
UGen300 8G / USB Talk, Reason, Automate

Local LLM assistant, RAG, an Ollama-compatible server and web UI — all on-device.



WQ1 • Meeting Notes

github.com/erp0917-stack/ugen300-demos



Local Python LLM

github.com/erp0917-stack/ugen300-demos



Local RAG Doc QA

github.com/erp0917-stack/ugen300-demos



hailo-ollama Server

blog.cavedu.com/.../asus-ugen300-usb-ai-ollama



Open WebUI Local Chat

blog.cavedu.com/.../asus-ugen300-usb-ai-open-webui

Ride the 2026 wave

hailo-ollama + Open WebUI + OpenClaw let customers use local AI like a cloud API — privacy stays on-device.

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17 Real Apps Which SKU Runs What

Every app ships with code. Model per app verified from the repo.

#	App	Model	Type	200	2G	8G	USB
01	Y1 Real-time Detection	yolov8m	Vision	✓	✓	✓	✓
04	Y2 Raise-hand to Flip	yolov8s_pose	Vision	✓	✓	✓	✓
07	Y4 Raise-hand Counter	yolov8s_pose	Vision	✓	✓	✓	✓
08	Y3 Air Instrument	yolov8s_pose	Vision	✓	✓	✓	✓
09	Y5 Yoga Coach	yolov8s_pose	Vision	✓	✓	✓	✓
10	Missing / Pack Check	yolov8m	Vision	✓	✓	✓	✓
13	Offline Voice Command	Whisper	Speech	—	✓	✓	✓
03	WQ1 Meeting Notes	Whisper+Qwen2.5-1.5B	Speech+LLM	—	△	✓	✓
12	Local Python LLM	Qwen2.5-1.5B	LLM	—	△	✓	✓
15	hailo-ollama Server	Qwen2.5 LLM	LLM	—	△	✓	✓
16	Open WebUI Local Chat	Qwen2.5 LLM	LLM	—	△	✓	✓
14	Local RAG Doc QA	Qwen LLM + retrieval	LLM	—	—	✓	✓
02	V1 Ask About Image	Qwen2-VL-2B	VLM	—	—	✓	✓
05	V2 Outfit Advisor	Qwen2-VL-2B	VLM	—	—	✓	✓
06	V3 Fridge Chef	Qwen2-VL-2B	VLM	—	—	✓	✓
11	FACE1 Face Report	Qwen2-VL-2B+Qwen2.5-1.5B	VLM+LLM	—	—	✓	✓
17	OpenClaw Remote Monitor	Qwen2-VL-2B	VLM+Agent	—	—	✓	✓

Bottom line: UGen200 runs 6 (all pure-vision, green). 13 need UGen300 — voice works from 2G; 8 need 8G (VLM / RAG / Agent).

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Source Code →  **GitHub**



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Seven Industries & Best-Fit SKU

Recommended models per vertical - vision on UGen200, GenAI on 8G.

Manufacturing

AOI defect · anomaly · safety

YOLOv8/11 · ResNet · PaddleOCR

UGen200 / 300-2G

Smart Retail

footfall · shelf audit · checkout

YOLO · CLIP · people-count

UGen200 / 300-2G

Security

intrusion · face · behavior · wildlife

Face det/rec · CenterPose · VLM

UGen200 → 8G (VLM)

Robotics

navigation · grasp · HRI

Depth · hand-landmark · pose

UGen300 M.2

Healthcare

imaging aid · fall · rehab score

UNet-seg · pose · zero-shot

UGen200 / 300-2G

Smart Traffic

vehicle · ALPR · parking

YOLO-OBB · PaddleOCR · tracking

UGen200 / 300-2G

Coding & Schools

AI/coding class · cram-school · interactive lessons

Pose · gesture · LLM tutor · Whisper

Any SKU · even old PCs

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What Products Can Be Used with the UGen Series?

Works with nearly all devices — except Apple's closed iOS ecosystem.



Raspberry Pi 5



Laptop



NUC



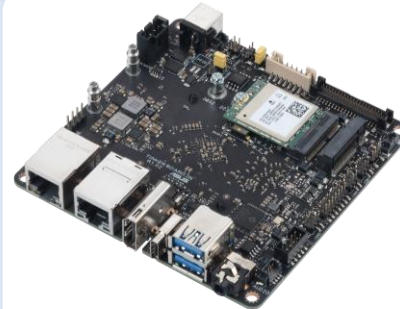
IPC



NAS



Mobile (Android)



SBC / Embedded



Desktop PC

Host requirements: x86 / ARM · Windows, Linux, Android · USB 3.1 Gen2 or PCIe M.2 slot

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UGen Series Development Flow

How to build your own AI app on UGen200 / UGen300 — five steps from plugging in to a running demo.

1 Plug in the UGen

UGen300 USB or M.2 (Hailo-10H), or UGen200 M.2 (Hailo-8L), on any Windows / Linux PC.

```
hailortcli scan
```

2 Install HailoRT

Driver + runtime + Python API, downloaded from the Hailo Developer Zone.

```
pip install hailort-*.whl
```

3 Get a model (.hef)

Download a pre-compiled model from the Hailo Model Zoo — or turn your own ONNX into .hef with the Hailo Dataflow Compiler.

```
yolov8m.hef
```

4 Load it in code

One line loads the .hef onto the accelerator; then feed it camera frames or prompts (see next page).

```
create_infer_model()
```

5 Build your app

Add camera, UI and your own logic — every demo shown today follows this pattern.

```
OpenCV + PySide6
```

UGen200 · Hailo-8L

HailoRT 4.x · Hailo Model Zoo v2.x
Use the hailo8l .hef · vision models (detection, pose, OCR ...)

UGen300 · Hailo-10H

HailoRT 5.x · Model Zoo v5.x + GenAI Model Zoo
Use the hailo10h .hef · vision + LLM / VLM / Whisper

UGen Series Code — One Line Loads the Model

Point the code at a .hef file and the model runs on the accelerator — everything after that is your app.

```
vision_demo.py · UGen200 / UGen300
1 from hailo_platform import VDevice, FormatType
2 import cv2, numpy as np
3
4 vdevice = VDevice()
5 model = vdevice.create_infer_model("yolov8m.hef") 1 load the model
6 model.output().set_format_type(FormatType.FLOAT32)
7 runner = model.configure()
8
9 img = cv2.resize(cv2.imread("desk.jpg"), (640, 640))
10 out = np.empty(model.output().shape, np.float32)
11 b = runner.create_bindings()
12 b.input().set_buffer(img)
13 b.output().set_buffer(out)
14 runner.run([b], 10000) 2 run on the chip
15 # out = detected objects: draw boxes, count people ...
```

```
chat_demo.py · UGen300 (Hailo-10H)
1 from hailo_platform import VDevice
2 from hailo_platform.genai import LLM
3
4 vdevice = VDevice()
5 llm = LLM(vdevice, "Qwen2.5-1.5B-Instruct.hef") 1 load the model
6
7 chat = [{"role": "user", "content": [{"type": "text",
8     "text": "Explain edge AI in one line."}]]
9 ans = llm.generate_all(prompt=chat, 2 generate
10     max_generated_tokens=100)
11 # ans = the model's answer: show it in your app UI
```

UGen200 runs the vision code with a hailo8l .hef. LLM / VLM from the GenAI Model Zoo need UGen300 (Hailo-10H).

1

Load — change the .hef name to switch models (e.g. yolov8s_pose.hef)

2

Run — feed camera frames or chat messages to the accelerator

3

Use — draw boxes, count people, show the answer: that's your app

Thank you for your Time.

All 17 apps ship with open-source code (MIT) and a trilingual developer handbook.

Resources

Demos: github.com/erp0917-stack/ugen300-demos

OpenClaw: github.com/Asus-UGen-Series/ugen300-openclaw-edge-remote-monitor

Guides: blog.cavedu.com · YouTube: ASUS UGen300 x OpenClaw / RPi5 / ZimaBoard

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APPENDIX

14 cinematic scenario stories — pain point to solution, full original pages.

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From Camera to JSON in 5 Seconds

Offline AI Object Detection, Plug-and-Play, Stable and Efficient

1 Capture

Use a phone to capture the objects in the scene



2 Local Inference

Connect ASUS UGen300 USB and run offline AI inference on a laptop



3 Output

Generate object annotations and a JSON summary in 5 seconds



Pain Point:

Network-dependent demo workflows carry high on-site risk



Edge AI:

Even offline, it can complete object annotation and statistical results



Stable Demo,
No Fear of Disconnects



Plug-and-Play
USB-C Connection



100% Offline
No Network Required



Local AI Inference
Privacy and Security



JSON Output
Structured Results



Plug-and-Play Offline Video Analysis

ASUS UGen300 USB AI Accelerator



100% Local
Processing



Your Data,
Your Control



No Cloud
Upload



Challenge: Surveillance footage cannot be uploaded to the cloud, and PoC validation takes time



Solution: Plug in UGen300 and the laptop instantly shows AI tracking results



30s Video → 1 Min Output



Input Video File_01.mp4

00:00:30

AI Tracking Analysis

- ✓ Detected People: 18
- ✓ Tracked Targets: 18
- ✓ Status: Completed

Analysis Progress

100%

Millisecond-Level Visual Impact You Can See

On-device inference and real-time detection to unleash true hardware performance

15–30 FPS Real-Time Inference

< 100ms Latency



On-Device Inference

Run AI models locally—no internet required



Low Latency

<100ms end-to-end response



Secure & Private

Your data stays on your device



Plug & Play

One USB-C connection, instant start

ASUS UGen300

USB AI Accelerator



Edge AI
Acceleration



Low
Latency



Secure &
Private



Plug & Play



Cloud drawback: higher latency, customers don't feel TOPS specs



On-device demo: point the camera and detect instantly, showing true hardware power



See Instant Performance.

Demo on-site. Performance you can trust.

Training-Free Zero-Shot Classification

Enter Custom Labels, Classify Instantly



Traditional Pain Point:

Requires Large Image Sets and Weeks of Training



Long Effort · High Cost



CLIP Zero-Shot Classification: Accurate Results in 5 Seconds

Enter Custom Labels

Normal Operation

Equipment Anomaly

Missing Safety Helmet



5-Second
Classification



Classification Results



Normal Operation 92.1%



Equipment Anomaly 6.3%



Missing Safety Helmet 1.6%

⚡ Local Inference Completed (1.24s)



5-Second
Classification
Ultra Fast



On-Device
Inference
Data Stays Local



Plug & Play
Simple Setup



Accurate
Classification

High Precision · Low False Alarms



Rapid
Deployment

Plug In · Start Instantly



Privacy &
Security

On-Device · Data Secure



Flexible
Scalability

Supports Various Models

ASUS UGen300
USB AI Accelerator⁴⁷

Missing Items on the Line? AI Catches Them Instantly



Manual reinspection is time-consuming and tiring



AI vision compares in real time and alerts on anomalies instantly



Missing Item



Inspection Complete

AI Vision + ASUS UGen300 USB

A real-time, accurate, and reliable solution for missing-item inspection on production and packaging lines.



Edge AI Acceleration

Local real-time inference, fast response



USB-C Plug and Play

Easy setup, fast deployment



Stable and Reliable

Built for long operating hours

UGen300 USB / Accelerating Edge Inference⁸

Smart Front-Door Protection

Local AI judgment, precise awareness of important moments



Filter Noise Alerts

Keep only person and package events



Telegram Two-Way Control

Remote viewing and instant commands



UGen300 USB Acceleration

Faster, more stable local AI inference



Runs 24/7 On-Prem

Reliable local processing around the clock



Notify Only What Matters



Person / Package Detection
Precisely detect targets



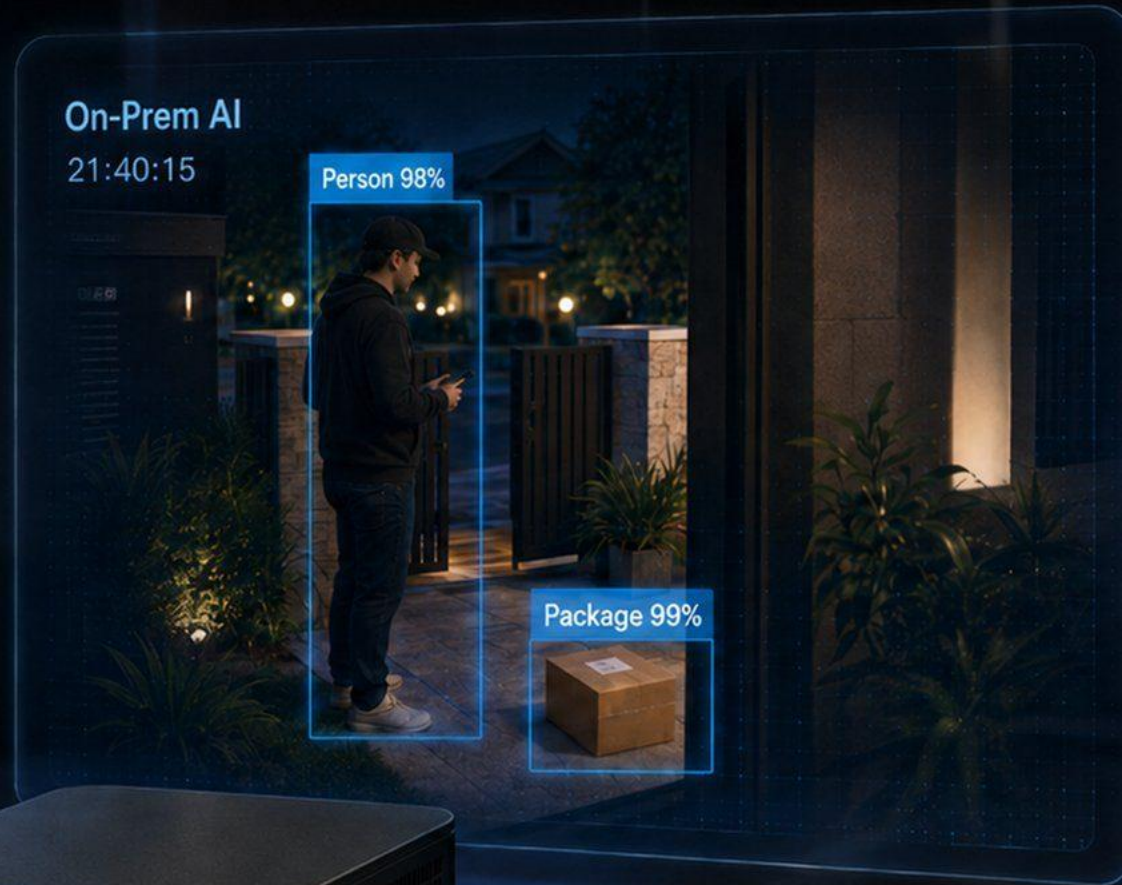
Smart Filtering
Reduce false alerts



Telegram Notification
Two-way interactive control



24/7 Protection
Runs locally 24/7



On-Prem AI

Telegram

now

Front Door AI

Person / Package detected

21:40



/ignore

21:40

Message



UGen300 USB Acceleration
Single USB-C Connection

Set a Confidential Zone in One Draw

Confidential-zone intrusion detection, fast alerts, flexible deployment



Traditional AI NVR is expensive
Self-built maintenance is difficult



Draw ROI with the mouse
Instant alert within 3 seconds



UGen300 USB Edge Deployment

Compact plug-and-play for any device, anywhere



No Server



No Complex Setup



Low Power &
Silent



Flexible
Deployment



Mouse-Drawn ROI
Flexible custom
detection zone



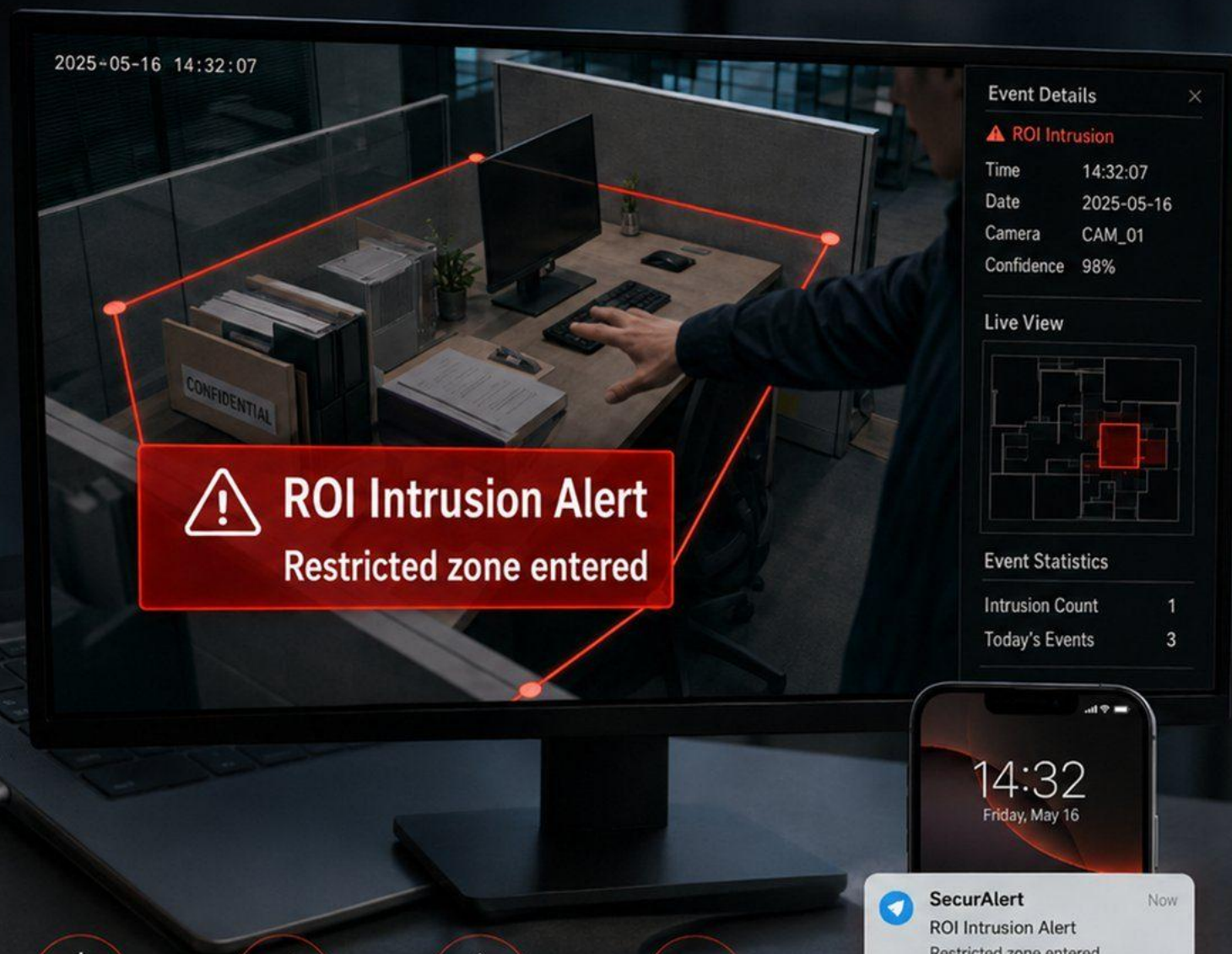
Alert Within 3 Seconds
Instant screenshot
push notification



Protect Confidential Zones
Block unauthorized
access



Flexible Edge Deployment
Compact, energy-saving,
plug and play



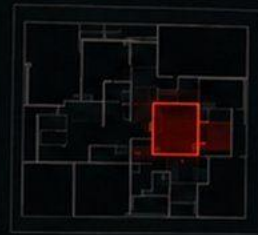
ROI Intrusion Alert
Restricted zone entered

Event Details

ROI Intrusion

Time 14:32:07
Date 2025-05-16
Camera CAM_01
Confidence 98%

Live View



Event Statistics

Intrusion Count 1
Today's Events 3



SecurAlert

Now

ROI Intrusion Alert
Restricted zone entered
Camera: CAM_01



Instant Screenshot Push

See Customer Flow, Protect Privacy.

UGen300 USB Flow Analytics,
Generate Daily LLM Insights.



Heatmap
Zones



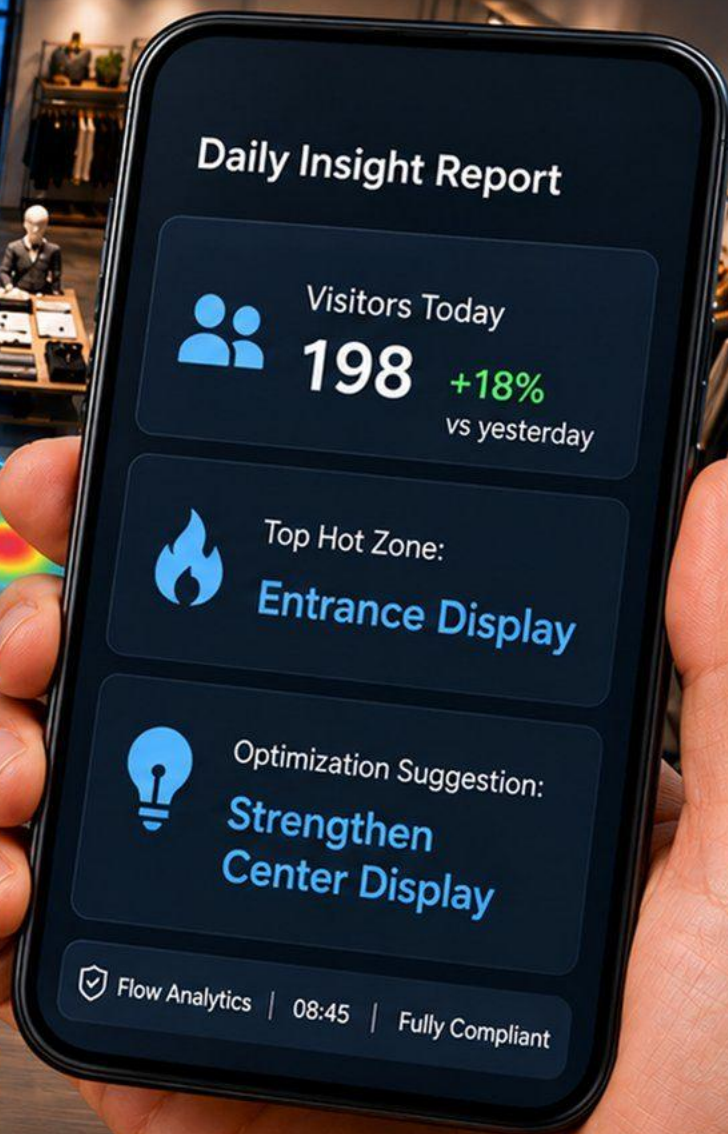
Traffic Flow
Tracking



No Video
Storage



On-device
Processing



Flow Analytics
Fully Compliant

The Ultimate Privacy Fall-Rescue Network



Dilemma:

Seniors resist installing cameras in private spaces, so falls are hard to discover in time

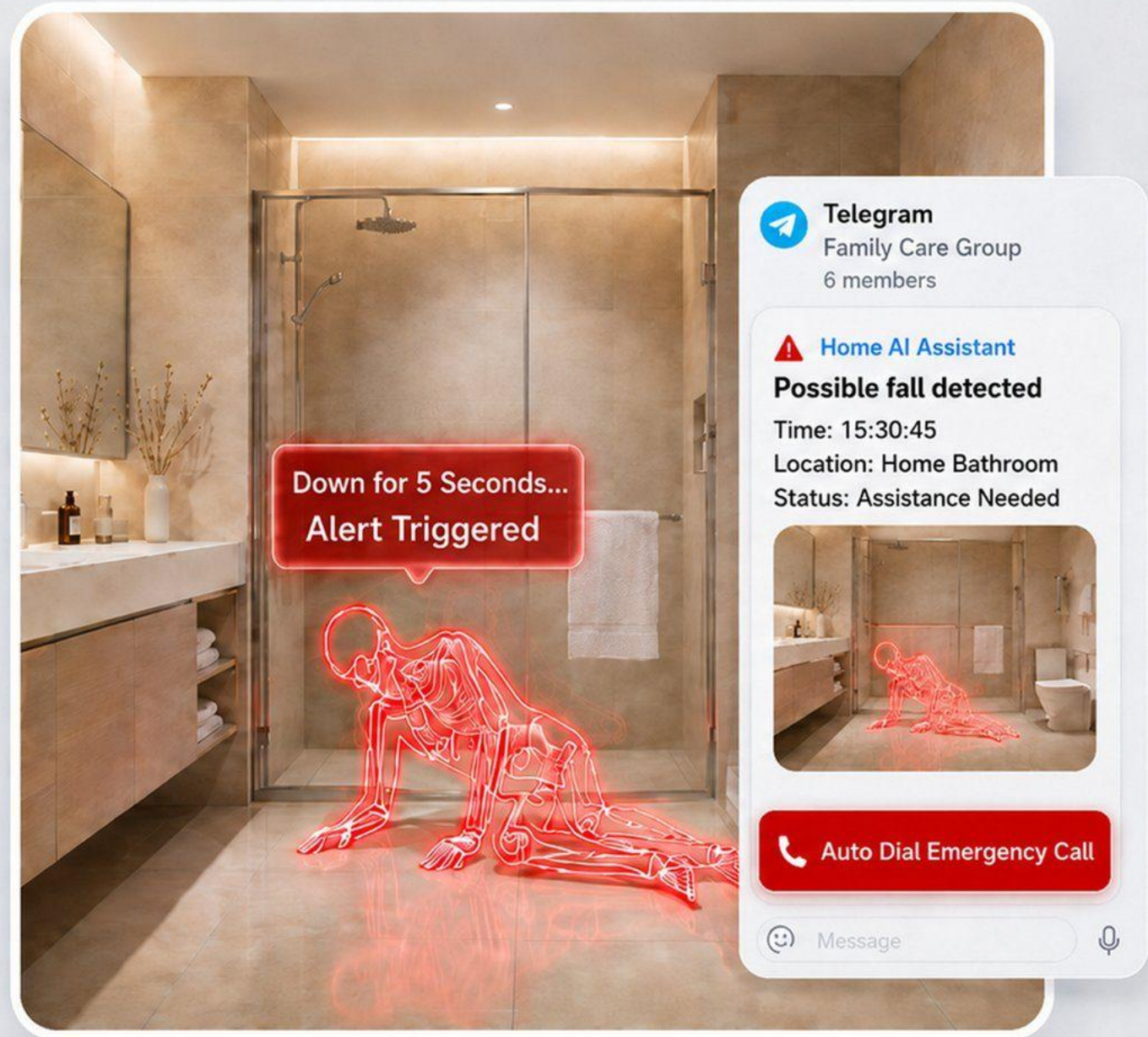


Skeleton Detection:

Only the body-outline skeleton is sent, preserving dignity so the family group can recognize the emergency and respond immediately



Local AI: Mini PC + ASUS UGen300 USB



A Warm Care Bot That Doesn't Feel Intrusive



Seamless Care



Natural Language Q&A



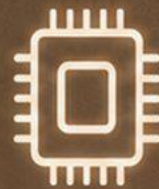
> 6 Hours Away Alert



Local Privacy Processing



Door Camera
(detects outings)



Local AI Inference
No upload, no recognition



Privacy Protection
Safe to use



ASUS UGen300
USB AI Accelerator

AI Counts So You Can Focus

✗ No more guessing
your rep count

✓ Precisely recognizes
each curl and updates
the count automatically



Live Feedback

AI Rep Counter



Take Back Control of Your Personal Privacy

Search photos locally without uploading to the cloud



Cloud Albums:

High monthly fees and higher risk of privacy leakage



Offline Search:

AI runs locally, and photos never leave your device



100% Local Processing,
Privacy Stays Off the Cloud

Free Your Hands, Command the Pace



Speakers Tied to Equipment:

Limited movement, weaker interaction



Gesture-Controlled Slides:

Smooth page turning, focus on what you say

<100ms
Low Latency

Market Insights

Industry Growth Trends



Key Metrics

+38%
Revenue
Growth

72%
Market
Share

+2.1x
Customer
Growth



Swipe Right: Next Page



Two Hands: Black Screen



ASUS UGen300 USB



The Digital Moat for Confidential Corporate Meetings

Local AI Processing × Cloud Upload Blocked × Live Transcription × Auto-Generated Decisions



Security Red Line

Legal restrictions on external reporting, cloud upload of recorded personal data, and manual note-taking is time-consuming



100%
Prevent Confidential
Business Data Leakage



Offline AI Engine

Live transcription + automatic Markdown meeting summary after the meeting



Local AI Processing

All data is processed only on-device, never leaving the device or the cloud



Cloud Upload Blocked

Block cloud transfer and eliminate leakage risk at the source



Live Transcription

High-accuracy AI speech recognition displays meeting content instantly



Markdown Meeting Summary

Automatically generate a structured summary after the meeting for clear decisions and action items



ASUS UGen300 USB AI Accelerator

The offline AI accelerator built for confidential meetings